Qutaiba W. Mantfaji

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SUMMARY

Mechanical Engineering student at AUIS with hands-on experience in embedded systems, analog circuit design, structural instrumentation, and fabrication. Holds OSHA and IOSH safety certifications. Eager to contribute to real-world engineering projects under professional mentorship.

EDUCATION

Bachelor of Science in Mechanical Engineering Feb 2024 – Present
American University of Iraq, Sulaimani (AUIS)

PROJECTS

Cantilever Beam Instrumentation System – ENGR 313: Measurement Systems, AUIS 2025

- Built a dual-sensor system using strain gauges (Wheatstone half-bridge) and MPU6050 accelerometer interfaced with Arduino Uno via I^2C.
- Implemented full static pipeline: strain → stress → bending moment → force, validating Euler-Bernoulli beam theory experimentally.
- Extracted natural frequency (5.83 Hz) via FFT on accelerometer data; results within ±15% of theoretical prediction.

Sawtooth Wave Generator – ENGR 390: Circuits, AUIS 2025

- Designed and built an analog op-amp circuit generating asymmetric sawtooth waveforms using a Schmitt trigger, integrator, and diode-based asymmetric RC network.
- Configured Schmitt trigger thresholds (±2.04 V), integrator (Rf = 100 kΩ, C = 100 nF), and asymmetry ratio (k ≈ 17.9) for target waveform.
- Implemented and tested on breadboard with dual ±12 V supply; verified output waveform against theoretical design.

EXPERIENCE

Product Designer – Makers of Baghdad Internship (On-site) Jul 2024 – Sep 2024

- Operated CNC and Faber Laser Machines for precision cutting, engraving, and component fabrication.
- Provided technical assistance and maintenance support in the fabrication lab.
- Managed material selection and cost optimization across multiple product designs.

CERTIFICATIONS

- **OSHA Oil and Gas Safety** (Participant) – Aug 2025 • **IOSH Working Safely** – Microteach, Jul 2025
- Rapid Prototyping Using 3D Printing Specialization – Arizona State University, Jan 2026
- Human Resources Management – Multisoft Systems, Sep 2025 • Business Model Canvas – Kennesaw State University, Jul 2025

TECHNICAL SKILLS

Proficient: Arduino, Sensor Integration (I^2C), Analog Circuit Design, CNC Laser Operation, Rapid Prototyping, 3D Printing
In Progress: SolidWorks, Fusion 360, AutoCAD, MATLAB, Minitab
Languages: Arabic (Native), English (Bilingual)

RELEVANT COURSES

Principles of Project Management • MATLAB for Numerical Analysis • Excel/Minitab for Engineering Statistics • Measurement Systems • Circuits